

GOVERNED KNOWLEDGE SYSTEMS · COMPANION SERIES · THESIS III

The Price of Recall

Shadow-Priced Cost Optimization for Contracted Retrieval
in Federated, Agent-Operated Systems

by

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A thesis submitted in partial fulfilment of the requirements
for the degree of

Master of Science in Computer Science

[Institution] · [Faculty / Department]

Supervisor: [Supervisor Name]

July 2026

Declaration

This manuscript is the third thesis of a companion series and, like its predecessors, a **design synthesis** presented to the conventions of a master's thesis. Its subject — the pricing and cost optimization of contracted retrieval — is notional: no measured implementation exists, no market has run, and every guarantee named in these pages is inherited from the literature as a *target shape*, never claimed as a result. The covenant it prices (Thesis II) and the fabric that hosts it (Thesis I) are internal specifications treated as fixed interfaces. Where a mechanism has an established basis — competitive caching, prophet and secretary bounds, knapsack relaxation, shadow-price rate control, differential privacy — the text marks it *adopt, not invent* and cites the source. The claimed contribution is the composition: one governed price serving a lexically floored contract system, with placement as capital, tails as obligations, and disclosure as a metered coupling.

Halifax, July 2026

Abstract

The retrieval covenant fixes the semantics of recall — what may be asked, what must be answered, what is refused aloud — but treats its economics as given: a single shadow price $\lambda(t)$, assumed to exist, rations compute and recall alike. This thesis asks where that price comes from, what it must aggregate, and how far a priced retrieval economy can be optimized before the price corrupts the practice it disciplines.

The cost of a retrieval is not a scalar: it compounds machine time, I/O and bandwidth, staleness risk, privacy expenditure ϵ across coupling channels, and the opportunity cost of budget consumed. The thesis contributes, first, a cost model mapping this vector into charter-weighted scalar form, with conservative rounding as a structural rather than procedural property — under uncertainty the system understates value and overstates cost, so ambiguity starves rather than floods. Second, an online pricing mechanism: shadow-price dynamics with explicit stability and fairness obligations, and admission analysed in knapsack and prophet regimes appropriate to bursty agent workloads. Third, a theory of pre-paid recall: caching, prefetching, replication, and stratum sizing recast as capital allocation — investments converting future retrieval cost into present admission cost under the same price, governed by the classical rent-to-buy discipline. Fourth, a tail-risk reserve exempting rare, high-consequence recalls from ordinary pricing, so cost optimization cannot silently forget what the federation can least afford to lose. Fifth, a federation chapter: prices as side channels, ϵ -metered price disclosure, and a standing Goodhart audit between metric and law. Six market invariants bind the design; a validation programme with pre-registered falsifiers — including the requirement that dynamic pricing beat a static tariff by a stated margin, or be retired — closes the series' arc: architecture, then covenant, then economy.

Keywords: retrieval economics; shadow pricing; online admission; competitive caching; tail-risk reserves; privacy-metered federation; design science.

Contents

Declaration	ii
Abstract	iii
Contents	iv
List of Figures	vi
List of Tables	vii
Nomenclature	viii
1 • Introduction	1
1.1 Motivation	1
1.2 Problem statement	1
1.3 Research questions	1
1.4 Contributions	2
1.5 Methodology and evidentiary posture	2
1.6 Thesis outline	2
2 • Background and Related Work	3
2.1 The economics of information and computation	3
2.2 Shadow prices and their dynamics	3
2.3 Competitive analysis and the rent-to-buy discipline	3
2.4 Caching, locality, and freshness	4
2.5 Optimal stopping and constrained selection	4
2.6 The pathology of measures, and the parents	4
3 • The Cost of a Recall	5
3.1 The five components	5
3.2 The eight economic obligations	6
3.3 Traceability	6
4 • Scalarization under Charter	7
4.1 Weights are charter instruments	7
4.2 Handling incommensurability honestly	7
4.3 The conservative valve (E4, M2)	7
4.4 The lexical exemption (E6, M1)	8
5 • The Online Price	9

5.1 The controller	9
5.2 Stability as an obligation	9
5.3 Starvation-freedom (M4)	10
6 • Admission in Two Regimes	11
6.1 The batch regime	11
6.2 The stream regime	11
6.3 The consistency requirement	12
7 • Pre-Paid Recall: Placement as Capital	13
7.1 Four instruments, one decision	13
7.2 The break-even law	14
7.3 Depreciation and write-off	14
8 • The Tail Reserve	15
8.1 Why a price sheds the wrong traffic	15
8.2 Mechanics and governance (M5)	16
9 • Federation: Prices as Information	17
9.1 Local prices, no global auctioneer	17
9.2 Prices leak (M3)	17
9.3 Shared placement and fair attribution	18
9.4 The Goodhart audit (M6)	18
10 • Invariants and the Validation Programme	19
10.1 The six market invariants	19
10.2 Phase π_0 — Cost model on replay	19
10.3 Phase π_1 — The static baseline	19
10.4 Phase π_2 — The governed price	20
10.5 Phase π_3 — Capital, tails, federation	20
11 • Discussion and Conclusion	21
11.1 Answering the research questions	21
11.2 Limitations	21
11.3 Conclusion — closing the arc	22
Bibliography	23
Appendix A — Glossary	25

List of Figures

3.1 The cost funnel	5
5.1 The price control loop	9
6.1 Two admission regimes	12
7.1 The break-even discipline	13
8.1 The tail reserve	15
9.1 Federation pricing	17

List of Tables

3.1 Economic obligation traceability	6
6.1 Mechanisms and inherited guarantee shapes	12
10.1 Market invariants M1-M6	19
A.1 Glossary of pricing terms	25

Nomenclature

Symbol	Meaning
$c_{\square}(r)$	the cost vector of a recall r : compute, I/O/bandwidth, staleness risk, privacy ϵ , opportunity
w_T	tenant T 's charter-set weight vector; the lawful exchange rates among cost components
$c(r) = \langle w_T, c_{\square}(r) \rangle$	the scalarized cost, comparable to bids and admissible into the price
$\lambda(t)$	the shadow price of shared capacity; the object of this thesis
λ_{batch}	the marginal density at capacity in the batch (knapsack) regime
$u(t), u^*$	measured and target utilization; the controller's error signal is $u - u^*$
γ	controller gain, capped for proven damping
B, ρ	tenant budget envelope and the reserved (tail) share of it
$\Delta \text{value} / \text{cost}$	bid density; the admission statistic in both regimes
$\langle d, q_{\min}, b \rangle$	the covenant's demand triple (Thesis II), consumed unchanged
TTL	time-to-live on answers; the depreciation clock for seated capital
ϵ	differential-privacy budget, spent by coupled recall and by price disclosure alike
E1–E8	the eight economic obligations (Chapter 3)
M1–M6	the six market invariants (consolidated in Table 10.1)

1 · Introduction

The covenant consumes a price it never explains. This chapter makes the price the object of study and states what a lawful retrieval economy owes the contracts it serves.

1.1 Motivation

Every load-bearing test in the first two theses of this series leans on one symbol. The fabric's gates admit a query when its bid clears $\lambda(t)$; the covenant's clause 1 seals a binding on the same condition; the metabolic doctrine retrieves a datum when its value density clears the very same price. The symbol is doing three systems' worth of work, and nowhere in the series has it been anything but *assumed*: posted by an unexamined auctioneer, aggregating unstated components, moving by unstated dynamics. An economy whose price is unexamined is not governed; it is merely quantified. And a retrieval economy has a property that makes unexamined prices dangerous rather than merely untidy: the price decides, at machine cadence and without appeal, which questions get answered — which is to say it decides, cumulatively, what the federation knows. The price of recall is an epistemic instrument [21], and this thesis treats it with the care that status demands.

1.2 Problem statement

Precisely: **specify the cost model, pricing mechanism, capital discipline, tail protection, and federation rules under which a single shadow price can lawfully ration contracted retrieval** — where “lawfully” means: without pricing the lexical floors the parent architectures forbid from sale; without starving any tenant's floor-class traffic; without shedding the rare recalls that matter most; without leaking workload intelligence through the price itself; and without the metric displacing the practice it measures. Each “without” becomes an invariant, and each invariant a falsifier.

1.3 Research questions

RQ1. What does a recall cost — as a vector, attributably, completely? (Chapter 3.)

RQ2. By what authority and what arithmetic does the vector become a scalar comparable to bids? (Chapter 4.) **RQ3.** What price dynamics and admission rules

serve bursty, adversarial agent workloads, and with what guarantee shapes? (Chapters 5–6.) **RQ4.** How do placement capital, tail obligations, and federation

constraints bound the optimization? (Chapters 7–9.)

1.4 Contributions

C1 — The cost model. A five-component cost vector with eight economic obligations (E1–E8), and a charter-governed scalarization whose conservatism is structural (Chapters 3–4). **C2 — The governed price.** A damped shadow-price controller with explicit stability and starvation-freedom obligations, and admission analysed in batch-knapsack and stream-threshold regimes with inherited guarantee shapes (Chapters 5–6). **C3 — The capital discipline.** Caching, prefetch, replication, and stratum sizing unified as rent-to-buy investment under the same price, with staleness as depreciation (Chapter 7). **C4 — The protective machinery.** A governed tail reserve, ϵ -metered price disclosure, a standing Goodhart audit, six market invariants, and a validation programme whose first falsifier retires the whole mechanism if it cannot beat a static tariff (Chapters 8–10).

1.5 Methodology and evidentiary posture

Design science, as before [11, 18], with one added discipline specific to an economics thesis: **guarantee hygiene**. The online-algorithms and stopping-theory literatures supply bounds — competitive ratios [20, 3], prophet and secretary constants [12, 9] — that hold under stated assumptions this design cannot always meet. Every such bound is therefore imported as a *target shape*: the form the validation programme tests against, never a property claimed by inheritance. All counts and thresholds are illustrative; the parents are fixed interfaces; and the registry discipline of the prior theses applies to the price's touchpoints.

1.6 Thesis outline

Chapter 2 reviews six literatures. Chapter 3 decomposes the cost of a recall and states obligations E1–E8. Chapter 4 gives the scalarization under charter. Chapter 5 specifies the price controller; Chapter 6 the two admission regimes. Chapter 7 develops placement as capital; Chapter 8 the tail reserve; Chapter 9 federation pricing. Chapter 10 consolidates the invariants and the validation programme; Chapter 11 discusses, concludes, and closes the series' arc.

2 · Background and Related Work

Six literatures meet in the price: information economics, computation markets, competitive analysis, caching, optimal stopping, and the pathology of measures.

2.1 The economics of information and computation

That acquiring information is a costly activity with diminishing returns is the founding observation of information economics [21]; that machine time itself could trade in a market was proposed within the decade [23] and matured into working computational economies [25]. Auction theory contributed the discipline of incentive-compatible mechanism design [24]. This thesis inherits the tradition's central move — decentralize allocation through a price — and departs from it at one point the tradition treats as heresy: some allocations (the lexical floors) are exempted from the price entirely, because the parent architectures rank obligations above efficiency.

2.2 Shadow prices and their dynamics

The rigorous foundation for prices as decentralized admission signals is network rate control: utilization-driven shadow prices with provable stability and fairness properties [14]. Classical feedback theory supplies the damping vocabulary the controller chapter uses [1]. Both are adopted, not invented; the composition — a Kelly-style price serving a hard-real-time admission test [15] over anytime kernels [26] — is the design's own.

2.3 Competitive analysis and the rent-to-buy discipline

Online decisions against unknown futures are the subject of competitive analysis [20, 3], and its most portable lesson is the rent-to-buy rule: commit capital only after observed demand has paid the capital price once over, achieving bounded regret against hindsight [13]. Chapter 7 makes this the law of every placement instrument.

2.4 Caching, locality, and freshness

Replacement has an unattainable optimum [2], a behavioural foundation in locality [7], an adaptive practice [17], and an empirical demand law — Zipf-like popularity — that makes small caches disproportionately valuable [4]. Freshness scheduling under change supplies the depreciation model: a seated answer loses value on a clock [5], which the covenant's TTL already measures. The capital chapter composes all four.

2.5 Optimal stopping and constrained selection

Batch selection under budget is the knapsack problem, whose fractional relaxation yields the density-threshold rule and, with it, the marginal price [6]. Irrevocable selection from streams is optimal stopping: secretary-style calibration [9] and prophet-style threshold guarantees [12] give the stream regime its shape. Chapter 6 requires the two regimes to agree on replay — a consistency condition the literatures, living apart, never needed to state.

2.6 The pathology of measures, and the parents

Any metric made a target deteriorates as a metric [22, 16] — for a pricing thesis this is the central occupational hazard, and Chapter 9 institutionalizes the response as a standing audit. Privacy accounting [8] governs what the price may reveal across tenants; cooperative attribution [19, 10] governs how joint gains from shared placement are credited. The parents fix the interfaces: the fabric supplies gates, budgets, membrane, and ledgers; the covenant supplies the demand triple, the stamps, and invariant V5 — one price, both banks — which this thesis must now make true rather than assume. The gap: no surveyed system prices contracted, floor-protected, federated retrieval under a single governed shadow price with tails exempted and disclosure metered. That composition is the thesis.

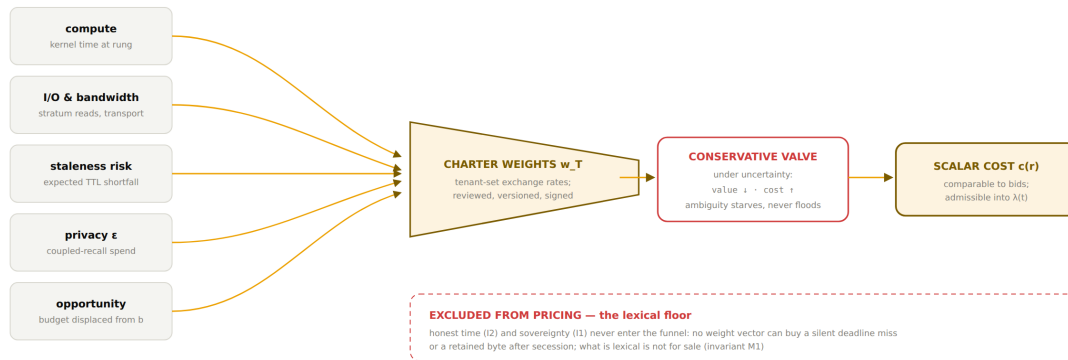
3 · The Cost of a Recall

Before a price can be honest, the cost must be complete: five components, eight obligations.

3.1 The five components

A recall r consumes: **compute** — kernel time at the bound rung, measured, not estimated, wherever the kernel is instrumented; **I/O and bandwidth** — stratum reads and transport, the dominant term for archive-class bindings; **staleness risk** — the expected shortfall between the answer's TTL and the demand's use horizon, a probabilistic cost the freshness literature prices on a clock [5]; **privacy ϵ** — the differential-privacy budget spent when the binding crosses the membrane [8], a strictly finite resource and therefore a cost, not a formality; and **opportunity** — the share of the demand's budget b this recall displaces from alternatives, the shadow-value term that makes the vector self-referential in exactly the way a price must resolve. Together: $c(r)$.

The cost of a recall is a vector; the charter, not the operator, decides how it becomes a number



the funnel is governance, not plumbing: weights are charter instruments, and changing them is a countersigned event

Figure 3.1: Five cost components enter; charter weights scalarize; the conservative valve rounds against admission; and the lexical floors never enter the funnel at all. The funnel is governance, not plumbing.

3.2 The eight economic obligations

E1 · Measurability. Each component is measured where instrumentation exists and conservatively modelled where it does not; the boundary between the two is declared per component. **E2 · Attribution.** Cost is booked to the covenant that caused it — no pooled slush, no socialized overrun. **E3 · Completeness.** No unbooked externality: staleness and ε are inside the vector precisely because they are the two costs systems habitually externalize. **E4 · Structural conservatism.** Under uncertainty, value is understated and cost overstated — as an estimator property, not an operator habit. **E5 · Comparability.** The scalarization renders costs commensurate with bids, or the admission test is theatre. **E6 · Lexical exclusion.** Floors are not for sale: honest time and sovereignty never enter the funnel. **E7 · Transparency asymmetry.** Within a tenant, the price and its components are fully inspectable; across tenants, disclosure is metered (Chapter 9). **E8 · Auditability.** Every booked cost re-derives from logged measurements under independent audit, in the registry discipline the series established.

3.3 Traceability

Table 3.1: The eight obligations and their discharging constructs.

E	Obligation	Discharging construct	Bound by
E1	measurability	component instrumentation map (Sec. 3.1)	M2
E2	attribution	per-covenant booking (covenant RC-13)	M2
E3	completeness	five-component vector incl. staleness, ε	M1, M2
E4	structural conservatism	the conservative valve (Sec. 4.3)	M2
E5	comparability	charter scalarization (Ch. 4)	M2
E6	lexical exclusion	floor exemption (Sec. 4.4)	M1
E7	transparency asymmetry	disclosure metering (Ch. 9)	M3
E8	auditability	re-derivation audit of bookings	M6

4 · Scalarization under Charter

The vector becomes a number by an act of governance, not an act of arithmetic.

4.1 Weights are charter instruments

The map $c(r) = \langle w_T, c(r) \rangle$ requires exchange rates among incommensurables: how much compute equals how much staleness equals how much ε . No operator has the authority to set those rates, because they encode what a tenant values — a biomedical estate weights staleness brutally; a litigation estate weights ε . The weights therefore live in the charter, versioned and countersigned like floors and consents, and changing them is a governance event the provenance chain records. The funnel of Figure 3.1 is drawn as governance for exactly this reason.

4.2 Handling incommensurability honestly

Weighted scalarization presumes the components trade smoothly at the margin, which is false at extremes: no quantity of cheap compute compensates an exhausted ε budget. The model therefore composes the linear funnel with per-component hard caps inherited from the charter — inside the caps, trade at w_T ; at a cap, the component becomes temporarily lexical and the recall must be refused or re-planned. This two-regime structure keeps the convenient arithmetic where it is honest and abandons it exactly where it would lie.

4.3 The conservative valve (E4, M2)

Every estimator in the funnel rounds against admission: value estimates take their lower confidence bound, cost estimates their upper. The valve is structural — built into the estimator interfaces, not entrusted to callers — because a rounding convention enforced by discipline erodes under load precisely when it matters. The systemic consequence is deliberate: ambiguity starves rather than floods, and the burden of proof for spending shared capacity lies with the spender.

4.4 The lexical exemption (E6, M1)

Two obligations never enter the funnel. Honest time: a deadline miss cannot be bought, so no weight exists against which to price it — lateness is breach, not expense. Sovereignty: retained bytes after secession, unconsented coupling, unmetered disclosure are not costs but prohibitions. The exemption is invariant M1 — *floors are not for sale* — and it is what distinguishes this economy from the market-first archetype the fabric thesis already examined and rejected: there, everything trades; here, the tradeable is bounded by the constitutional.

5 · The Online Price

A governed control loop whose mandate is boring prices: responsive enough to ration, damped enough never to become the story.

5.1 The controller

The price moves by utilization error: $\lambda' = \gamma (u - u^*)$, with per-resource, per-stratum sensing, a target u^* set below saturation to preserve deadline headroom, and gain γ capped so the closed loop is provably damped under the workload models the twin world can replay [14, 1]. Multiplicative variants and per-resource prices are admissible refinements; what is not admissible is discretion — the controller's parameters are charter-visible, and its trajectory is a logged, auditable time series.

The shadow price as a governed control loop — stability and fairness are obligations, not accidents

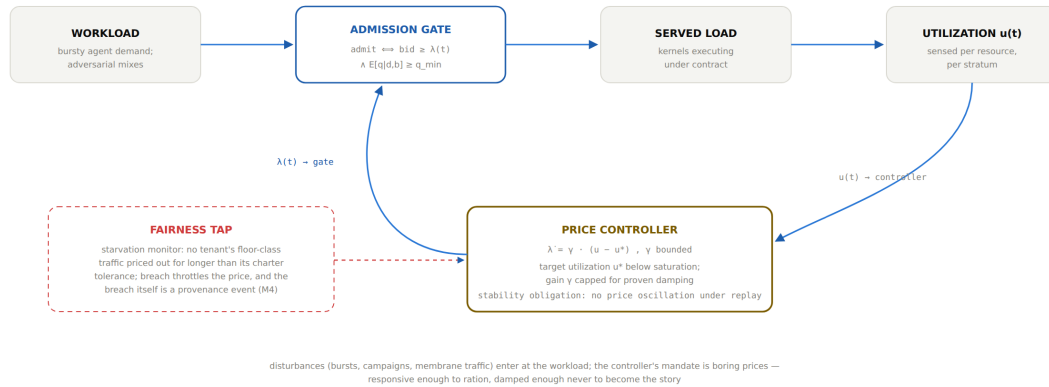


Figure 5.1: The shadow price as a governed feedback loop. Utilization drives the price; the price gates admission; the fairness tap can throttle the price when floor-class traffic starves; every excursion is a provenance event.

5.2 Stability as an obligation

Price oscillation is not an aesthetic defect: an oscillating λ whipsaws the admission gates, converts marginal contracts into refusal storms, and teaches tuners to game timing rather than value. The stability obligation is therefore falsifiable and blunt: under the validation programme's replayed and adversarial loads, the price exhibits no sustained oscillation and settles within declared bands after bursts. A controller that cannot meet the band is retuned or replaced; the mandate, stated once in the figure and repeated here, is *boring prices*.

5.3 Starvation-freedom (M4)

A pure price starves the poor lawfully. The fairness tap forbids it: per-tenant, per-class starvation monitors track how long floor-class demand has been priced out; when a charter-set tolerance is exceeded, the tap throttles the price for that class — an explicit, logged intervention, not a hidden subsidy. M4's design honesty is that the intervention is visible: the federation can see exactly when and for whom efficiency yielded to obligation, which is what distinguishes a floor from a favour.

6 · Admission in Two Regimes

Batch selection for campaigns, threshold stopping for streams — one calculus, two clocks, and a consistency requirement between them.

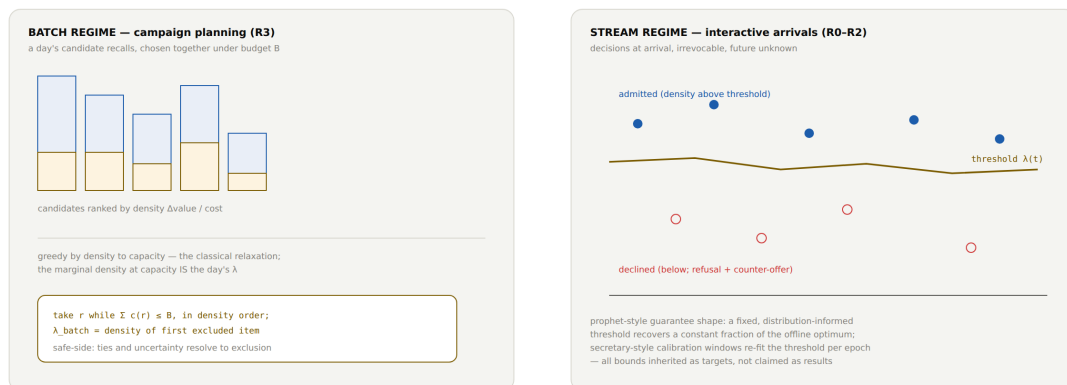
6.1 The batch regime

Campaign planning (rung R3) sees its candidates together: a day's proposed archive sweeps, coupled recalls, and consolidations, each with estimated value and scalarized cost, selected under the period budget. This is knapsack, and the fractional relaxation's density-threshold rule [6] does two jobs at once: it selects (take in density order to capacity, with safe-side resolution of ties and uncertainty to exclusion), and it *prices* — the density of the first excluded candidate is the period's marginal price, λ_{batch} , the cleanest empirical estimate of what capacity is worth that the system possesses.

6.2 The stream regime

Interactive rungs decide at arrival, irrevocably, against an unknown future — optimal stopping, not optimization. The design adopts the threshold shape: a distribution-informed density threshold, refit per epoch from ledger history in secretary-style calibration windows [9], with prophet-style constant-fraction guarantees as the inherited target [12]. Guarantee hygiene applies with full force: agent workloads are neither i.i.d. nor exchangeable in general, so every constant is a shape to test on replay, never a theorem about this system.

Two admission regimes: batch knapsack for campaigns, threshold stopping for streams



one calculus, two clocks: the batch λ and the stream λ must agree on replay — divergence is a falsifier (Chapter 10)

Figure 6.1: Batch knapsack for campaigns; threshold stopping for streams. The marginal density at capacity and the online threshold are two estimates of one quantity — their agreement on replay is a falsifier, not a hope.

6.3 The consistency requirement

λ_{batch} and the stream threshold estimate the same economic quantity on different clocks. If, on replayed workloads, they diverge persistently beyond their declared band, then the system is running two economies under one symbol — precisely the fork the covenant's one-price invariant (V5) forbids between banks, reappearing between regimes. The consistency requirement is therefore promoted to a standing monitor and a falsifier (Chapter 10), and the guarantee-shape table is given below.

Table 6.1: Adopted mechanisms, their regimes, and the guarantee shapes imported as validation targets — none claimed as results.

Mechanism	Regime	Guarantee shape (target)	Source
greedy by density	batch (R3 campaigns)	optimal for the fractional relaxation; near-optimal integral under small items	[6]
threshold stopping	stream (R0–R2)	constant-fraction of offline optimum under prophet assumptions	[12]
calibration window	stream, per epoch	1/e-style exploration bound under secretary assumptions	[9]
rent-to-buy placement	capital (Ch. 7)	2-competitive against hindsight	[13]
adaptive replacement	cache baseline	self-tuning recency/frequency balance	[17]
damped price control	continuous	stability under bounded gain	[14, 1]

7 · Pre-Paid Recall: Placement as Capital

Caching, prefetching, replication, and stratum sizing are one decision — buy future retrievals now — governed by one break-even law.

7.1 Four instruments, one decision

The covenant's duality already taught that storage is pre-paid retrieval; this chapter prices the pre-payment. A **cache seat** buys avoided descents for as long as the seated answer's TTL lives; a **prefetch** buys deadline headroom at the risk of forecast waste; a **replica** buys transport and tail latency at the cost of storage and coherence; a **stratum expansion** buys systematic density gains at generation timescale. All four are claims on the same budget, admitted through the same auction, at the same λ — capital allocation is admission with a longer horizon, and Zipf-shaped demand [4] is why small, well-chosen commitments dominate.

Pre-paid recall: placement decisions are investments priced by the same λ

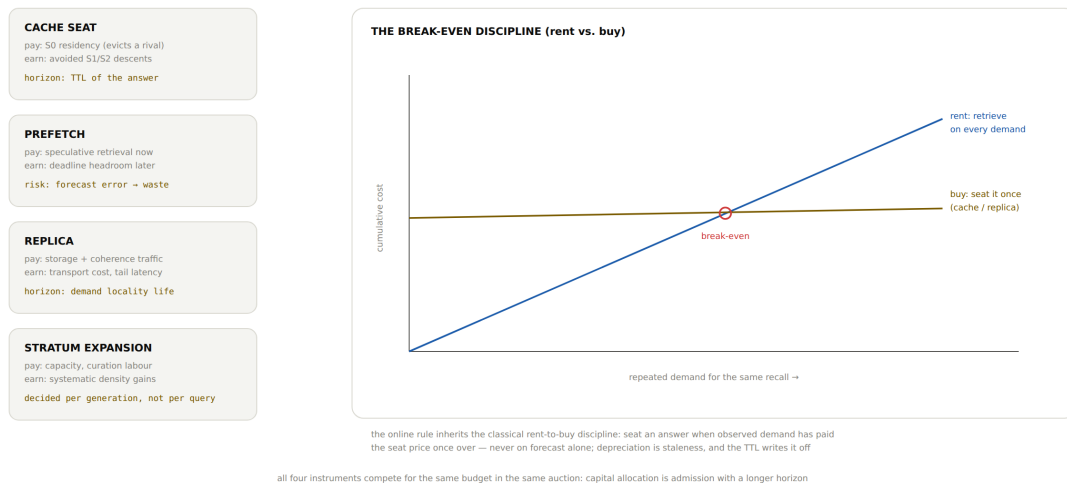


Figure 7.1: Rent versus buy. The online rule seats capital only after observed demand has paid the seat price once over — bounded regret against hindsight; staleness is depreciation and the TTL writes it off.

7.2 The break-even law

Every instrument obeys the rent-to-buy discipline [13]: commit only after observed demand has paid the commitment price once over — never on forecast alone. The law's virtues are exactly what a governed economy needs: it is online (no oracle), bounded-regret (never worse than twice hindsight, as a target shape), and legible (the ledger shows the demand that justified every seat). Belady's optimum stands above the whole chapter as the unattainable benchmark replay can approximate [2], and adaptive replacement supplies the working baseline any learned policy must beat to earn deployment [17].

7.3 Depreciation and write-off

Seated capital decays: the answer's TTL is its depreciation schedule [5], and a seat whose TTL expires is written off automatically — no zombie capital, no stale answer served from a paid seat (the covenant's V1 would breach). Locality justifies the whole enterprise [7], and its failure is the chapter's honest risk: under adversarially non-local demand, every instrument underperforms rent, the break-even law correctly refuses to buy, and the estate simply pays as it goes — which is the correct degenerate behaviour, not a failure of the discipline.

8 · The Tail Reserve

Some recalls are obligations, and obligations do not bid.

8.1 Why a price sheds the wrong traffic

Under budget pressure a pure price system sheds exactly the traffic that pays least per unit cost — and in a retrieval economy that is, systematically, the tail: the rare provenance reconstruction, the safety-relevant archive sweep, the legally mandated audit recall. These pay rarely and matter absolutely; their value is realized in the seldom worlds where they are needed, which is precisely the signal a utilization-driven price cannot see. The metabolic doctrine met the same problem with a tail-event vault; this thesis meets it with the budget's mirror image: a reserve.

The tail reserve: what the federation can least afford to forget is exempt from the price

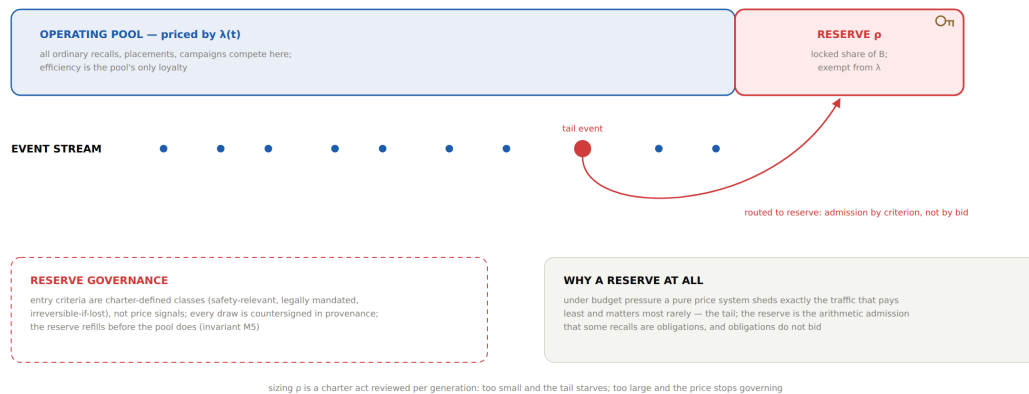


Figure 8.1: A locked share p of the budget, exempt from λ . Entry is by charter-defined criterion, never by bid; every draw is countersigned; the reserve refills before the operating pool does.

8.2 Mechanics and governance (M5)

The reserve ρ is a charter-set share of each budget envelope, locked away from the auction. Admission to it is by *criterion*, not by bid: charter-defined classes — safety-relevant, legally mandated, irreversible-if-lost — whose membership is checkable at admission and auditable after. Every draw is countersigned into provenance; the reserve refills before the operating pool when budgets renew; and sizing ρ is a per-generation charter act with a stated trade: too small and the tail starves, too large and the price stops governing. Invariant M5 — *the reserve is inviolable under load* — is the falsifiable core: no utilization level, no price excursion, no operator convenience may draw it down for ordinary traffic.

9 · Federation: Prices as Information

A tenant's price is a confidential measurement of its own operations; disclosure is a coupling and is metered like one.

9.1 Local prices, no global auctioneer

Each estate runs its own controller over its own utilization: $\lambda_A(t)$, $\lambda_B(t)$, $\lambda_C(t)$. There is no global price because there is no global owner — the fabric's membrane doctrine, applied to economics. Cross-tenant recalls (the covenant's R3 coupled column) clear at the *supplier's* price plus membrane costs, so a tenant's capacity is never rationed by a price it does not govern.

Prices in a federation are information: disclosure is metered like any other coupling

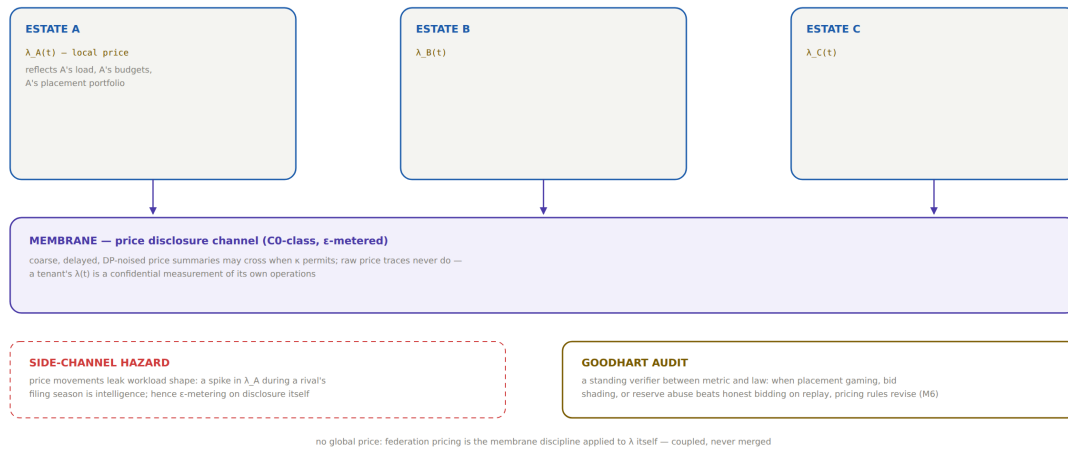


Figure 9.1: Local prices per estate; coarse, delayed, DP-noised summaries may cross the membrane under consent; raw price traces never do. The Goodhart audit stands between metric and law.

9.2 Prices leak (M3)

A price trace is a workload fingerprint: a spike in λ_A during a rival's filing season is competitive intelligence, and a correlated dip reveals a shared dependency. Disclosure is therefore itself a priced, consented coupling: coarse, delayed, DP-noised summaries may cross the membrane under κ with the ϵ -meter debited [8]; raw traces never cross. Within the tenant, by contrast, the price is fully transparent (E7) — the asymmetry is deliberate and constitutional.

9.3 Shared placement and fair attribution

Consortium shards and jointly funded replicas create joint gains with no natural owner. Attribution follows the series' established answer: Shapley-valued contribution accounts [19, 10], now denominated in avoided retrieval cost, so that a tenant's credit for a shared seat is its marginal contribution to the coalition's savings. Free-riding becomes legible in the same ledger that prices everything else.

9.4 The Goodhart audit (M6)

A pricing system is a standing invitation to optimize the measure [22, 16]: bid shading against the controller's lag, placement gaming against the break-even rule, reserve-criterion stretching, threshold timing. The audit is a permanent adversarial verifier — the gauntlet's economic twin — that replays ledger history searching for strategies that beat honest bidding; any it finds are published to the federation and the pricing rules revised through the parents' governed evolution machinery. M6's phrasing is the series' oldest discipline in new clothes: an audit must stand between metric and law.

10 · Invariants and the Validation Programme

Six market invariants, four phases, and the falsifier that can retire the whole mechanism.

10.1 The six market invariants

Table 10.1: The consolidated invariants, their enforcement, and their falsifiers.

M	Invariant	Enforcement	Falsified by
M1	floors are not for sale	lexical exemption (Sec. 4.4)	any priced path to a floor violation found in audit
M2	conservatism is structural	valve in estimator interfaces (Sec. 4.3)	ambiguous bids observed flooding rather than starving
M3	transparent within, metered across	disclosure channel + ϵ -meter (Sec. 9.2)	workload reconstruction from disclosed summaries
M4	no lawful starvation	fairness tap (Sec. 5.3)	floor-class starvation beyond charter tolerance
M5	the reserve is inviolable	criterion admission + countersigned draws (Sec. 8.2)	any ordinary-traffic draw under load
M6	audit between metric and law	standing Goodhart verifier (Sec. 9.4)	a published gaming strategy left unremedied

10.2 Phase π_0 — Cost model on replay

Steps. Instrument the five components on the twin world's replayed workloads; book costs per covenant; run the re-derivation audit. **Burden.** E1-E3 demonstrated: attribution complete, externalities booked, audit exact. **Exit.** A costed ledger the later phases can price against.

10.3 Phase π_1 — The static baseline

Steps. Implement the null hypothesis: a two-tier static tariff (peak/off-peak) with the same floors and reserve. Measure served value, refusal rates, and tail retention on replay. **Burden.** The baseline is tuned in good faith — the comparison is honest only if the tariff is given its best case. **Exit.** A defensible benchmark and the declared margin the dynamic mechanism must beat.

10.4 Phase π_2 — The governed price

Steps. Bring the controller live in the twin; run the batch and stream regimes; arm the consistency monitor; inject bursts and adversarial mixes. **Burden.** Stability bands held; the M4 tap engages correctly under engineered starvation; λ_{batch} and the stream threshold agree within band. **Exit.** The central falsifier evaluated: **if dynamic pricing fails to beat the static tariff by the declared margin on served value at equal floor compliance, the mechanism is retired** and the series records that a tariff suffices — an honourable result the programme must remain willing to publish.

10.5 Phase π_3 — Capital, tails, federation

Steps. Enable placement under the break-even law against the adaptive baseline; lock the reserve and attack it with load; open metered disclosure between twin tenants; run the Goodhart audit with planted gaming strategies. **Burden.** Placement beats rent-only and the adaptive baseline on replay [17]; M5 holds under every engineered pressure; planted strategies are detected and remedied; tail retention does not degrade as budgets tighten. **Exit.** The economy operates end-to-end in the twin with all six invariants green — the state in which live deployment could even be discussed.

11 · Discussion and Conclusion

Answers, limits, and the close of the arc: architecture, covenant, economy.

11.1 Answering the research questions

RQ1: a recall costs five measured or conservatively modelled components, booked per covenant under obligations E1-E8 (Chapter 3). **RQ2:** the vector becomes a scalar by charter-owned weights inside hard caps, through a structurally conservative valve, with the lexical floors exempted entirely (Chapter 4). **RQ3:** a damped, utilization-driven controller prices admission in two regimes — density-threshold batch and stopping-theoretic stream — whose agreement is monitored, with all guarantee shapes imported as targets (Chapters 5-6). **RQ4:** placement is rent-to-buy capital under the same price; tails are criterion-admitted obligations outside the price; and federation pricing is local, metered in disclosure, fairly attributed, and audited against Goodhart (Chapters 7-9).

11.2 Limitations

The guarantee shapes are inherited from literatures whose assumptions — independence, exchangeability, known distributions — agent workloads will violate; the validation programme tests shapes, but a passed replay is not a theorem. The charter-weight model presumes tenants can articulate exchange rates they may only discover through regret. The staleness and opportunity components are the vector's softest terms, and E1's measured/modelled boundary will be contested exactly there. Federation pricing assumes the membrane's ϵ accounting composes across price disclosure and coupled recall — plausible under standard composition [8], unproven in this setting. And the series' single-designer bias culminates here: an economy invites adversaries more inventive than its author, which is why M6 is an institution rather than a chapter.

11.3 Conclusion — closing the arc

The series set out to give autonomous, federated knowledge systems a constitution in three movements. The fabric fixed the *habitat*: sovereign estates, a graded membrane, a deadline spine, and a design that only governed experience may rewrite. The covenant fixed the *meaning* of the habitat's central act: retrieval as a bilateral contract with derived warrants, purchased depth, and one economy across memory and search. This thesis fixed the *economics* of that act: a complete and attributable cost, a governed and boring price, capital under a break-even law, tails outside the market, disclosure behind the meter, and an audit standing permanently between metric and law. Knowing, forgetting, and paying for both now share one accountable frame — and every load-bearing claim in it carries a falsifier, including the one that would retire the price itself. The arc closes where the programme's oldest rule began: nothing here asks to be believed; everything asks to be tested.

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Appendix A — Glossary

Table A.1: Terms of art used throughout the thesis.

Term	Definition
break-even law	commit capital only after observed demand has paid the commitment price once over (Ch. 7)
charter weights w_T	tenant-owned exchange rates among cost components; changing them is a countersigned event
consistency monitor	the standing check that λ_{batch} and the stream threshold agree on replay (Sec. 6.3)
conservative valve	the estimator-level rule: understate value, overstate cost; ambiguity starves (Sec. 4.3)
cost vector $\vec{c}(r)$	compute $\cdot$ I/O/bandwidth $\cdot$ staleness risk $\cdot$ privacy ϵ $\cdot$ opportunity (Sec. 3.1)
depreciation	the decay of seated capital on the answer's TTL clock; expiry is automatic write-off
disclosure metering	DP-noised, delayed price summaries across the membrane; raw traces never cross (Sec. 9.2)
fairness tap	the starvation monitor empowered to throttle the price for floor-class traffic (Sec. 5.3)
Goodhart audit	the standing adversarial verifier searching for strategies that beat honest bidding (Sec. 9.4)
guarantee hygiene	importing literature bounds as validation targets, never as claimed results (Sec. 1.5)
lexical exemption	honest time and sovereignty never enter the funnel; floors are not for sale (M1)
λ_{batch}	the marginal density at capacity in campaign planning; the cleanest empirical price estimate
reserve ρ	the locked budget share admitting recalls by criterion, not by bid; inviolable under load (M5)
static tariff baseline	the null hypothesis the dynamic mechanism must beat by a declared margin, or retire (Sec. 10.3)